# MARANDA 2025 KCSE PREDICTION

## EXPECTED EXAM 1

# ELECTRICITY

**448/2**

**PAPER 2 (Practical)**

**TIME: 2½ HOURS**

**NAME……………………………………………………………………………………**

**SCHOOL……………………………………………………… SIGN…………………………**

**INDEX NO…………………………………………………… ADM NO……………………**

***Kenya Certificate of Secondary Education.***

## INSTRUCTIONS TO CANDIDATES

- **a.** *Write your **Name**, **Class** and **Index Number** in spaces provided*
- **b.** *There are **five** exercises in this examination*
- **c.** *Candidates are allowed **30** minutes for each exercise*
- **d.** *Each exercise will be awarded a maximum of 20 marks*
- **e.** *At each station, candidates are **not allowed** to either view the previous station's work or read instructions for the other stations*
- **f.** *Attempt all the **four** exercises*
- **g.** *All dimensions are in **millimeters** unless otherwise stated*
- **h.** *Candidates should answer the questions in **English***

**For examiner's Use only**

| EXERCISE | 1 | 2 | 3 | 4 | TOTAL |
|---|---|---|---|---|---|
| MARKS(EC) | | | | | |
| MARKS(MC) | | | | | |
| GRAND TOTAL | | | | | |

## EXERCISE 1

Using the materials, equipment and measuring instruments provided, perform the following tasks

**a)** Connect the circuit as shown in **Figure 1** **($2\frac{1}{2}$ marks)**

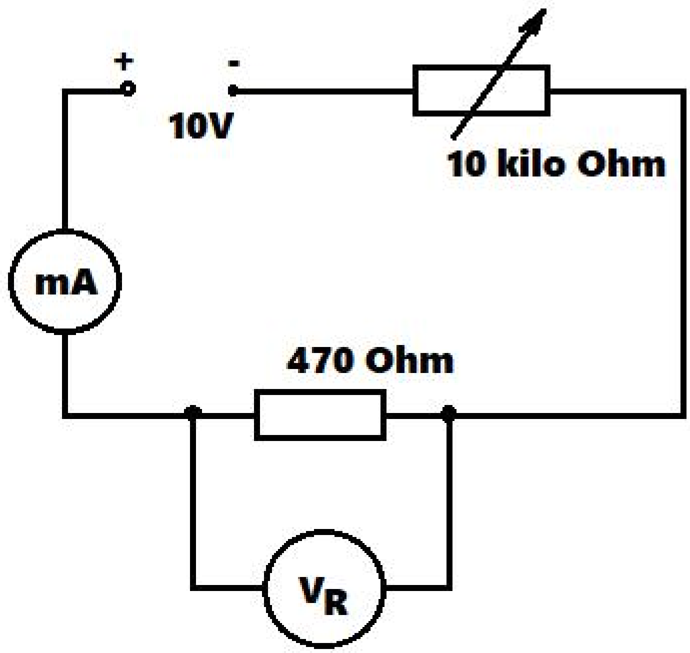



**Figure 1.**

**b)** Connect the circuit to the power supply **(1 marks)**

**c)** Turn the power on and adjust the potentiometer to obtain each of the voltages ($V_R$) shown in **Table 1**.in each case read and record the corresponding current I. **(6 marks)**

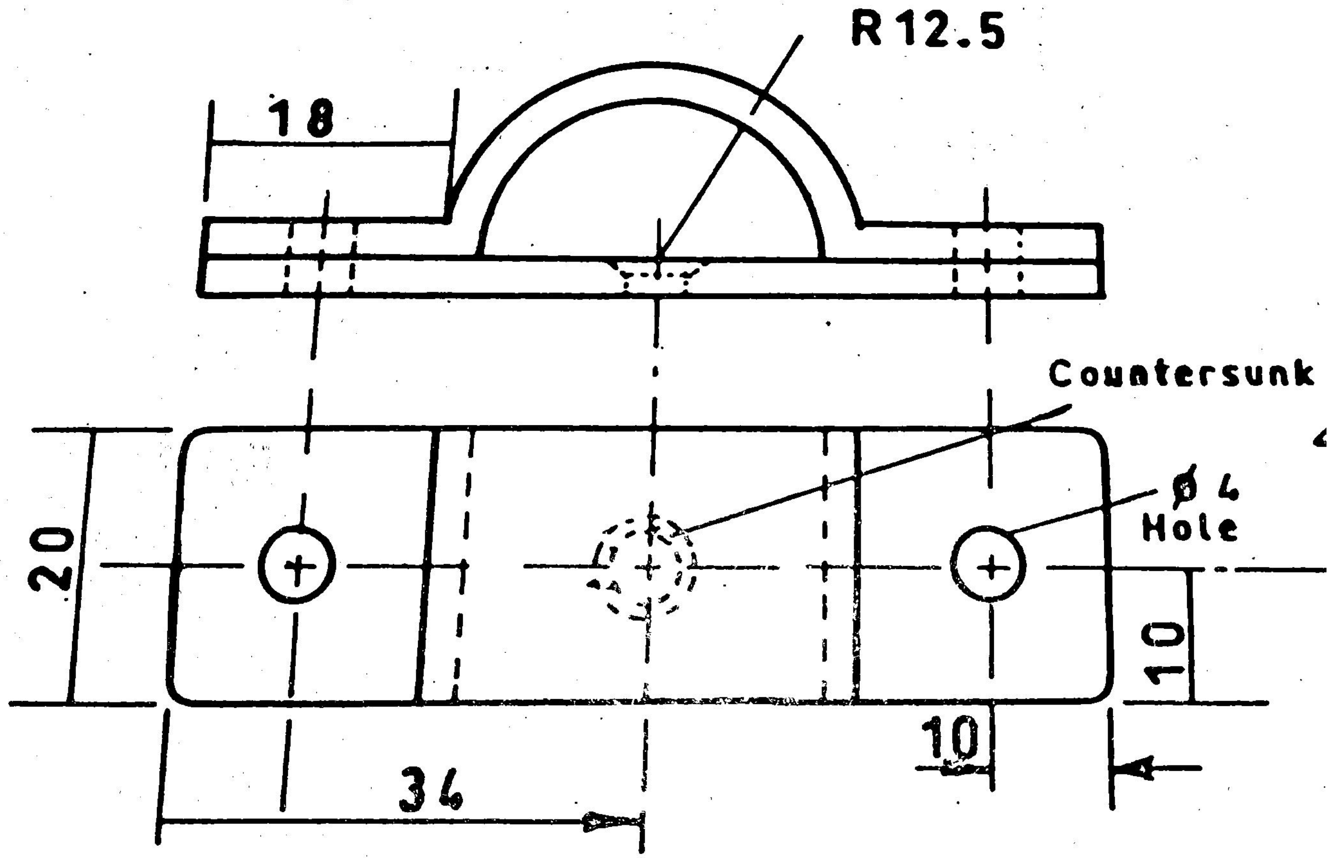
| $V_R$ (Volts) | I(mA) | Power (mW) |
|---|---|---|
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |
| 8 | | |

**d)** For each value of VR, calculate the power dissipated in the 470Ω resistor and complete the table **(5 marks)**

**e)** Using the values in **Table 1** draw the graph of power against voltage $V_R$. **($5\frac{1}{2}$ marks)**

## EXERCISE 2

Using tools equipment and materials provided make the space bar saddle as shown in **Figure 2**




## EXERCISE 3

**a.** Connect circuit as shown in **figure 3**. Let examiner check your work (2 marks)

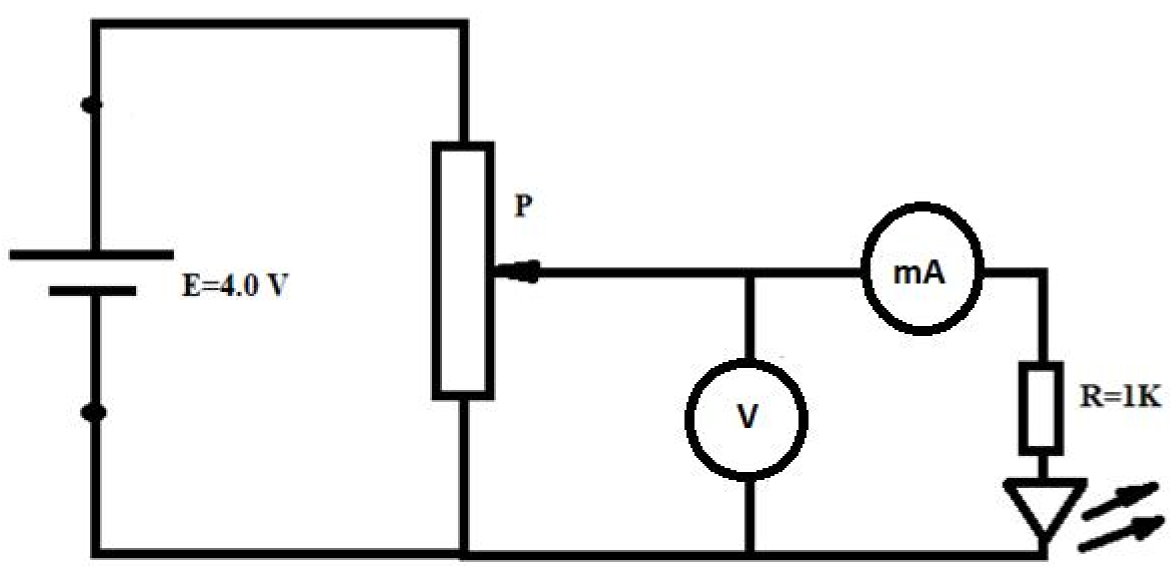


**Figure 3**

**b.** Adjust the potentiometer and complete the table below (6 marks)

| V(v) | 0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 |
|---|---|---|---|---|---|---|---|---|
| I(mA) | | | | | | | | |

**c.** Adjust the potentiometer until LED starts to glow. Record voltage___________ **(1 mark)**

**d.** Plot graph of I(mA) against V(v) **(10 marks)**

EXERCISE 4

Figure 4 shows a layout of a power final sub circuit. Using PVC conduit wiring system, install the circuit

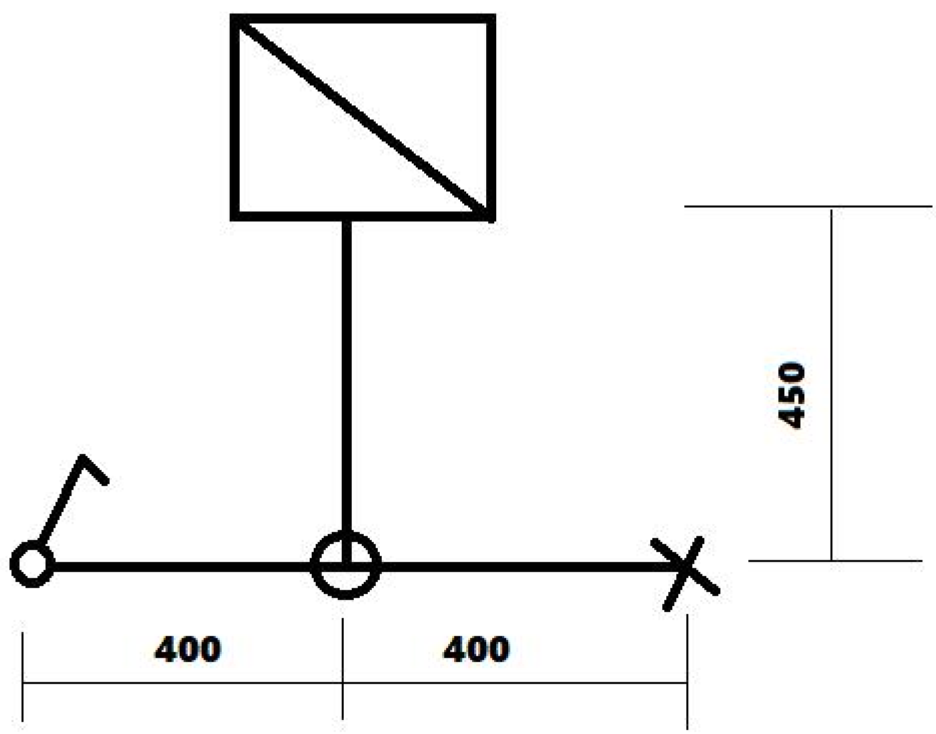



Figure 4